

Let $\mathbf{r}$ be the separation vector from a fixed point (x', y', z') to the point (x, y, z) , and let r be its length. Show that

(a) $\nabla(r^2) = 2 \mathbf{r}$

(b) $\nabla\left(\frac{1}{r}\right) = -\hat{\mathbf{r}}/r^2$

(c) What is the general formula for $\nabla(r^n)$?

Solution:

$$\mathbf{r} = (x - x', y - y', z - z')$$

$$r = \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}$$

(a)

$$\begin{aligned}\nabla(r^2) &= \nabla\left((x - x')^2 + (y - y')^2 + (z - z')^2\right) = \\ &= 2(x - x')\hat{\mathbf{x}} + 2(y - y')\hat{\mathbf{y}} + 2(z - z')\hat{\mathbf{z}} = 2 \mathbf{r}\end{aligned}$$

(b)

$$\begin{aligned}\nabla\left(\frac{1}{r}\right) &= \nabla\left(\frac{1}{\sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}}\right) = \\ &= -\frac{1}{2} \frac{2(x - x')\hat{\mathbf{x}} + 2(y - y')\hat{\mathbf{y}} + 2(z - z')\hat{\mathbf{z}}}{\sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}^3} = -\frac{\mathbf{r}}{r^3} = -\frac{\hat{\mathbf{r}}}{r^2}\end{aligned}$$

(c)

$$\begin{aligned}\nabla(r^n) &= \nabla\left(\left((x - x')^2 + (y - y')^2 + (z - z')^2\right)^{\frac{n}{2}}\right) = \\ &= \frac{n}{2}(2(x - x')\hat{\mathbf{x}} + 2(y - y')\hat{\mathbf{y}} + 2(z - z')\hat{\mathbf{z}})\left[(x - x')^2 + (y - y')^2 + (z - z')^2\right]^{\frac{n-2}{2}} = \\ &= nr^{n-2} \mathbf{r}\end{aligned}$$

□